



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

- Q. 2 Answers the following short questions: (2×10=20)
- (a) Define the term microstate.
 - (b) What are effective collisions?
 - (c) Give two statements of 2nd law of thermodynamics.
 - (d) Give a mathematical relation between Gibbs energy, entropy and enthalpy of a process.
 - (e) What is the effect of temperature on vertical distribution of gas?
 - (f) Describe term Clausius in-equality.
 - (g) Write two different forms of Arrhenius equation.
 - (h) What is difference between Classical and Statistical thermodynamics?
 - (i) Give mathematical formulation of Barometric formula.
 - (j) What do you understand by fast reactions?
- Q.3 (a) State and explain 2nd law of Thermodynamics. (5)
 (b) Discuss Clausius in-equality in Thermodynamics. (5)
- Q.4 (a) Give five postulates of transition state theory. (5)
 (b) Derive Eyring equation on the basis of postulates of transition state theory (5)
- Q.5 (a) What is Sterling's approximation? Give its significance. (5)
 (b) Discuss effect of molar mass and altitude on vertical distribution of particles. (5)



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q. 2 Answer Short Questions.

(10x2=20)

- a) What is closed process?
- b) What is partition function?
- c) Define second law of thermodynamics.
- d) What is meant by degree of freedom?
- e) Describe most probable velocity.
- f) What is cage effect?
- g) Reactions in solution phase are faster than in gas phase. Why?
- h) What is law of conservation energy?
- i) What is difference between spontaneous and non-spontaneous processes?
- j) Explain Nernst heat theorem.

Section III (Questions with Brief Answers)

(3 x 10 = 30)

1. Define third law of thermodynamics and give its experimental verification. (10)
2. Derive and explain Maxwell's Distribution Law for distribution of molecular velocities among gas molecules. (10)
3. (a) Discuss the effect of temperature on the rate of reaction on the bases of Arrhenius equation. (5)
(b) Derive Eyring equation using postulates of transition state theory. (5)



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short notes on following.

(15x2=30)

- (a) What are ionic reactions?
- (b) Describe barometric formula?
- (c) What is cage effect?
- (d) How can you determine pre-exponential factor from Arrhenius equation?
- (e) What is the effect of temperature on rate constant?
- (f) What are effective collisions?
- (g) Give postulates of Bimolecular theory?
- (h) Define partition function? What is factorization of partition function?
- (i) Describe Nernst Heat Theorem.
- (j) Write down Nernst Approximation formula.
- (k) What is the difference between classical and statistical thermodynamics?
- (l) What is Clausius inequality?
- (m) What do you understand by fast reactions?
- (n) Derive root means square velocity from Maxwell's distribution law?
- (o) What is pulse method?

Answer the following questions.

- Q.2 (a) What is Eyring equation? How can you determine Eyring parameters using Eyring equation? (5)**
- (b) What is the effect of temperature and altitude on vertical distribution constant? (5)**
- Q.3 (a) Derive expression of energy in terms of "Partition Function". (5)**
- (b) Verify third law of thermodynamics experimentally. (5)**
- Q.4(a) Elaborate five postulates of transition state theory? (5)**
- (b) Derive two Maxwell's relations? (5)**